

R Cheat Sheet : PREDICTION (1)

Data Analysis (ISBMM104 - UNamur)

Dataset: `data`
Outcome: `outcome`
Predictors: `pred1`, `pred2`, `pred3`

USEFUL PACKAGES

<code>library(glmnet)</code> <i># LASSO regression</i>	<code>> cv.glmnet(), coef(), predict()</code>
<code>library(survival)</code> <i># Survival outcomes</i>	<code>> Surv()</code>
<code>library(pROC)</code> <i># AUROC</i>	<code>> roc(), auc()</code>
<code>library(rms)</code> <i># Calibration & bootstrap validation</i>	<code>> calibrate(), validate()</code>

PREPARE DATA

model.matrix() Creates the predictor matrix needed for LASSO.

```
x <- model.matrix(
  outcome ~ pred1 + pred2 + pred3,
  data = data
)[, -1]
```

`y <- data$outcome` Creates an outcome vector

Surv() Creates a survival outcome for Cox models.

```
y_surv <- Surv(data$time, data$event)
```

FIT MODEL

cv.glmnet() Fits a penalised regression model with cross-validation. Used for LASSO models.

```
lasso_logistic <- cv.glmnet(
  x = x,
  y = y,
  family = "binomial",
  alpha = 1
)
```

MODEL OUTPUT

coef() Shows model coefficients.
Coefficients equal to 0 are removed by LASSO.

```
coef(lasso_logistic, s = "lambda.min")
```

PREDICTIONS

predict() Generates predictions for each patient.

```
predicted_probability <- predict(
  lasso_logistic,
  newx = x,
  s = "lambda.min",
  type = "response"
)
```

PERFORMANCE & VALIDATION

roc() Creates the ROC object for binary outcomes.

```
roc_obj <- roc(
  response = data$outcome,
  predictor = predicted_probability
)
```

auc() Calculates the AUROC.

```
auc(roc_obj)
```

plot() Draws the ROC curve X-axis = 1 - Specificity
Y-axis = Sensitivity

```
plot(roc_obj)
```

validate() Bootstrap validation with rms models.

```
validate(
  model,
  method = "boot",
  B = 200)
```

calibrate() Creates bootstrap calibration estimates.

```
cal <- calibrate(
  model,
  method = "boot",
  B = 200)
plot(cal)
```